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Property-Conditioned Generative Design of Minibinders

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Abstract

Small designed proteins that bind a specific molecular target offer a promising route to new therapeutics, but practical candidates must combine target compatibility with suitable biochemical properties such as charge and hydrophobicity. Rather than treating these properties only as a filter after design, we seek to steer them directly during sequence generation. To do so, we adapt OmegAMP, a diffusion model originally developed to generate antimicrobial peptides with requested biochemical properties, for minibinder design. Our model generates amino acid sequences guided by both the desired biochemical properties and the proposed binder shape. We integrate our model into a pipeline that is capable of generating binders of desired properties given a target. Our in silico experiments show that the model is able to steer generation towards specified properties with standardized errors comparable to those reported for OmegAMP. The model is also able to follow its structural guidance and can recover the overall fold. Unfortunately, precise structural control remains challenging. Success in generating proteins likely to bind is largely limited to the easier setting of inpainting missing regions of partially specified amino acid sequences. Our work clarifies what can and cannot be achieved within the constraints of the general approach we set out to adopt. In particular, the model can rapidly generate proteins that satisfy the desired properties and broadly match the intended structure, making it useful as part of our pipeline, where unsuitable candidates can be filtered out. However, we believe that further refinement of the diffusion model alone is unlikely to yield reliable binder generation within the representational constraints adopted here.

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1 Introduction

Designing small proteins that bind to a specific molecular target offers a promising route to new therapeutics. For example, a small protein that binds to the SARS-CoV-2 spike protein can block the virus from entering cells (Cao et al., 2020). The task of designing such a minibinder is to find an amino acid sequence that produces a protein capable of the intended interaction. This depends on both the shape the protein adopts and the chemical features it presents to the target. Its shape determines how the two molecules can come into contact, while the chemistry of their surfaces determines the interactions those contacts can support.

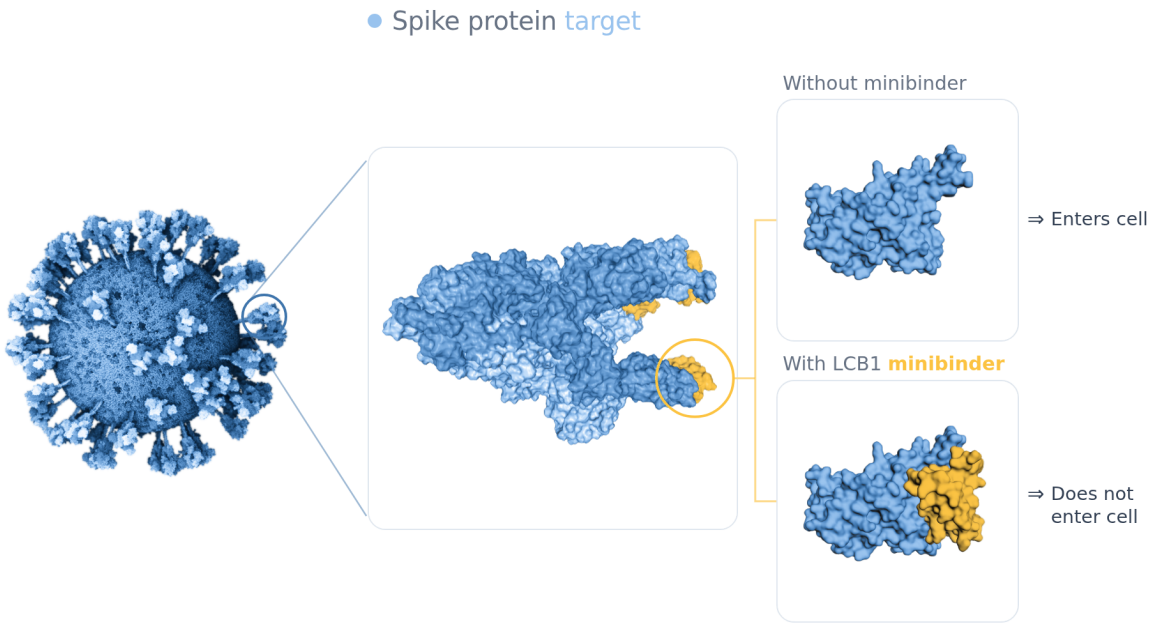


Figure 1. A minibinder targeting the SARS-CoV-2 spike protein. The designed minibinder LCB1 (gold) adopts a shape complementary to the viral spike protein (blue) and binds its receptor-binding region, preventing the virus from entering host cells.

However, the ability to bind does not by itself make a protein suitable for its intended use. Its biochemical properties also influence how it interacts with its surroundings. Charge influences electrostatic interactions with other molecules and can affect solubility, while hydrophobicity influences interactions with water (Lawrence et al., 2007). For example, exposed hydrophobic patches can cause binder molecules to stick together and form aggregates, reducing their availability to bind the intended target (Chennamsetty et al., 2009). A candidate may therefore be capable of binding the chosen target without being useful in practice. These considerations motivate control over the binder as a whole, rather than only the features directly responsible for its interaction with the target.

The difficulty is that these requirements cannot be satisfied independently. The amino acids that determine a protein's charge and hydrophobicity also contribute to its structure and to the chemical interactions it can form. Changing a sequence to obtain a different biochemical profile may therefore alter its shape or disrupt contacts needed for binding. Conversely, preserving the features required for binding can restrict the changes available to adjust its overall properties. It is consequently not enough to generate a sequence with the requested properties, nor to find a sequence compatible with the target and assume that its properties can be adjusted afterwards. The two requirements need to be considered together.

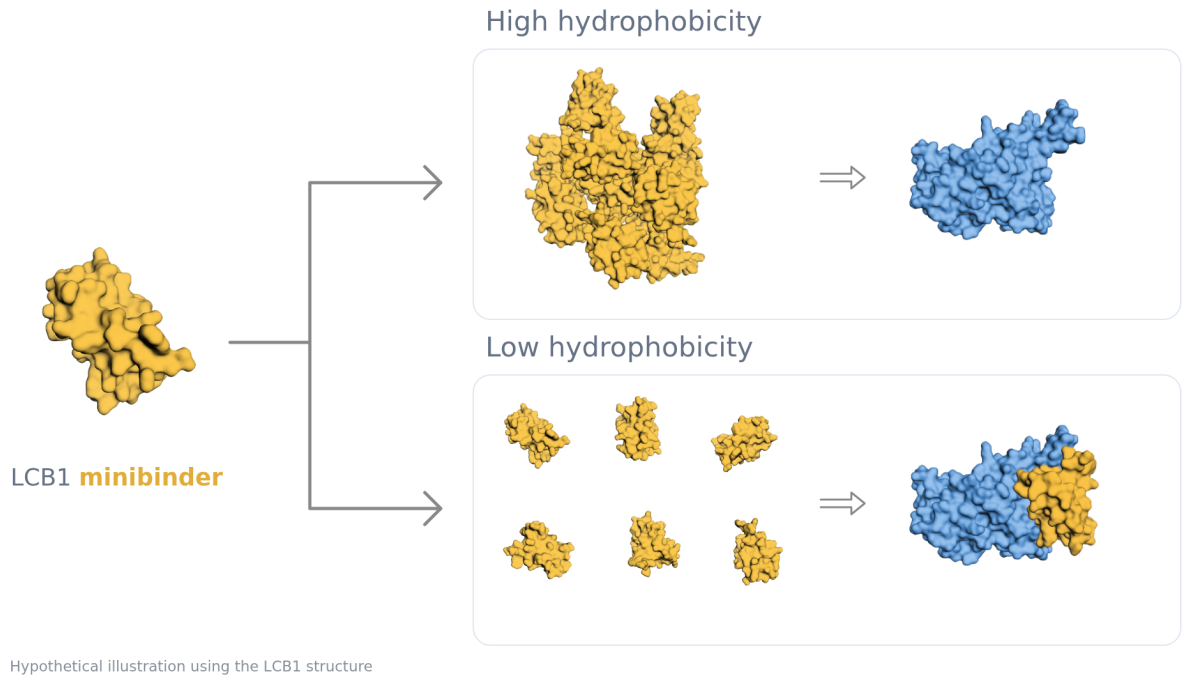


Figure 2. Effect of hydrophobicity on minibinder behaviour. High surface hydrophobicity can cause binder molecules (gold) to self-associate into aggregates rather than remaining available in solution to bind the target (blue).

Motivated by these two requirements, we treat target compatibility and biochemical properties as a joint design problem and develop a system for generating candidate minibinders. Given a chosen target and a requested property profile, the goal is to propose amino acid sequences that satisfy that profile while remaining compatible with the target. Rather than treating biochemical properties only as criteria for assessing candidates after generation, we seek to make them part of the design specification itself. This does not remove the need to assess the resulting candidates, but it allows the desired properties to influence which sequences are proposed in the first place.

To realize this approach, we build on OmegAMP, a diffusion model that generates antimicrobial peptides conditioned on requested biochemical properties (Soares et al., 2026). Because minibinder design also requires a shape complementary to the target, we extend OmegAMP so that sequence generation is guided by a proposed binder structure alongside the desired property profile. We place this generator within a pipeline that starts from a candidate binder shape for the chosen target, generates sequences designed to satisfy both the structural and biochemical specifications, and then evaluates the resulting candidates against the target.

To define the biochemical specification within this pipeline, we follow OmegAMP in focusing on length, net charge and mean hydrophobicity as a small set of properties to control. These provide a way to express requirements on the candidate as a whole without prescribing its complete amino acid sequence. They are not intended as an exhaustive description of what makes a binder suitable, nor do we seek a single profile that would be desirable for every application. Instead, the objective is to allow their desired values to be specified for an individual design task, both when generating a complete sequence from the candidate shape and when completing a partially specified sequence. In the latter setting, the supplied amino acids impose additional constraints, since the missing regions must be chosen so that the completed protein satisfies the overall requirements while retaining the specified sequence.

We evaluate this pipeline in silico by testing control over the requested biochemical properties, recovery of the supplied binder structure, and predicted compatibility with the target. Our experiments show that the model steers biochemical properties with accuracy comparable to OmegAMP and can recover the overall binder fold, while generating candidates confidently predicted to bind the target succeeds mainly when completing partially specified sequences rather than generating complete sequences from scratch.

The remainder of this report is organized as follows. We first review the background on OmegAMP and related methods for protein structure and binder design. We then describe how we formulate the conditioning descriptor, adapt the diffusion model architecture and training procedure, and assemble the complete design pipeline. Finally, we present our experimental evaluation of each stage and discuss the limitations of our approach alongside directions for future work. Practical details on where the model and pipeline are available and how to run them are provided in the appendix.

2 Background and Related Work

To control the biochemical properties of generated sequences, we build on OmegAMP, a model originally developed to generate antimicrobial peptides (AMPs), which are short chains of amino acids that inhibit or kill microorganisms and offer promising candidates for treating antibiotic-resistant infections (Soares et al., 2026). Many AMPs act by attaching to microbial surfaces and disrupting their lipid membranes. Because bacterial membranes typically present negatively charged surfaces, positive charges on a peptide favour attraction to these surfaces, while negative charges oppose it. In a simplified description, these contributions add together, so the total excess of positive over negative charge helps determine the peptide's electrostatic attraction to the membrane. Similarly, hydrophobic side chains favour contact with the nonpolar membrane interior, while hydrophilic side chains favour contact with the surrounding water, so their relative contributions influence the balance between the peptide's affinity for bacterial membranes and the aqueous environment (Dathe et al., 2001). Together with length, which influences the shapes and contacts a peptide can form, these properties provide useful design variables, although their relationship to activity depends on the peptide (Liu et al., 2007). OmegAMP exploits this to generate AMPs by steering its generation towards biochemical property profiles associated with antimicrobial activity (Soares et al., 2026).

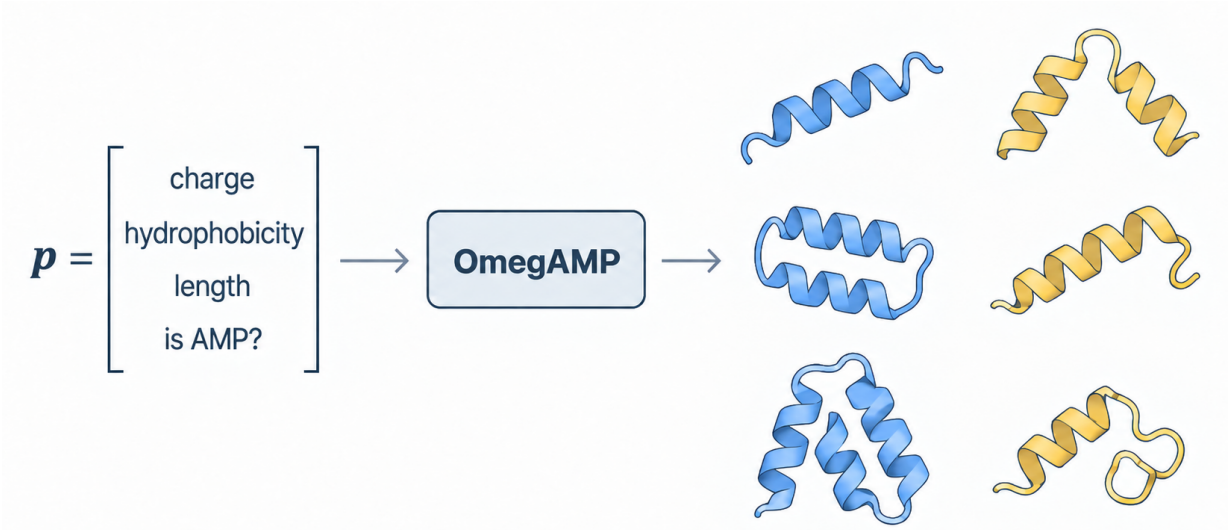


Figure 3. Property-conditioned peptide generation in OmegAMP. Given requested global properties such as net charge, hydrophobicity, and length, the model generates antimicrobial peptide candidates adopting diverse three-dimensional folds.

To achieve this control, OmegAMP uses a conditional diffusion model operating on a continuous representation of the amino-acid sequence. Each residue is encoded using numerical features derived from its biochemical properties, yielding an ordered array of residue features. During training, noise is added to these arrays and a U-Net learns to recover the clean representation. The peptide's global properties are supplied as additional inputs, teaching the model to reconstruct sequences consistent with those properties. To generate a new peptide, the model starts from random noise and repeatedly denoises it under the requested conditions. The resulting continuous features are then mapped back to amino-acid identities (Soares et al., 2026).

In this respect, OmegAMP differs from several prominent generative models for protein sequences, which model amino-acid identities rather than biochemical residue features. Autoregressive protein language models such as ProGen (Madani et al., 2023) generate one amino acid at a time from the preceding sequence, with additional annotations allowing generation to be directed towards particular protein families or functions. Generative adversarial networks such as ProteinGAN (Repecka et al., 2021) instead train a sequence generator against a discriminator that learns to distinguish generated sequences from natural ones. Even among diffusion models, EvoDiff (Alamdari et al., 2023) reverses the corruption of discrete amino-acid identities, whereas OmegAMP denoises continuous residue features to control global biochemical properties.

In binder design, however, we do not merely need to generate protein sequences with suitable biochemical properties, but sequences that also fold into a shape compatible with a chosen target. Whereas membrane-active AMPs interact with lipid bilayers, a protein binder must present an interface complementary to a particular target. Because OmegAMP only models sequences and their biochemical properties, adapting it for binder design requires drawing on structure-based binder design methods, which take a target structure as input and seek both a complementary three-dimensional binder shape and an amino-acid sequence that folds into it (Cao et al., 2022).

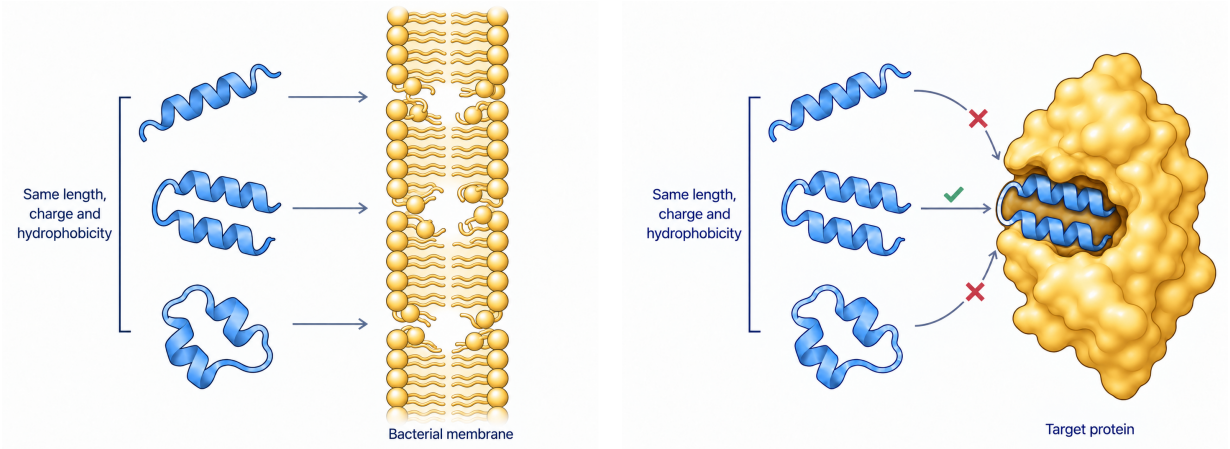


Figure 4. Why biochemical properties alone do not suffice for binder design. Peptides sharing the same length, charge, and hydrophobicity can all interact with a bacterial membrane despite adopting different folds (left), whereas binding a protein target requires a specific complementary three-dimensional shape (right).

Existing methods differ in whether they develop this sequence and structure together during initial design or first generate a backbone before designing the sequence. In the first approach, a single procedure produces or optimizes the binder's sequence and three-dimensional shape together for the target, whereas in the two-stage approach, one model first generates a binder shape that fits the target and a second model then designs an amino-acid sequence likely to fold into that shape, a step known as inverse folding. For example, Proteina-Complexa trains a generative model to directly output a binder's amino-acid sequence and three-dimensional structure given the target (Didi et al., 2026), while BindCraft uses a structure-prediction model, AlphaFold-Multimer, to predict how a randomly initialized sequence folds with the target and backpropagates a design objective to iteratively update the sequence before using ProteinMPNN to redesign non-interface positions (Pacesa et al., 2025). In a two-stage pipeline, a widely used combination pairs RFdiffusion with ProteinMPNN, where RFdiffusion iteratively denoises residue positions and orientations guided by the target to generate the binder structure (Watson et al., 2023), and ProteinMPNN uses a graph neural network to generate an amino-acid sequence one residue at a time from that structure's geometry (Dauparas et al., 2022).

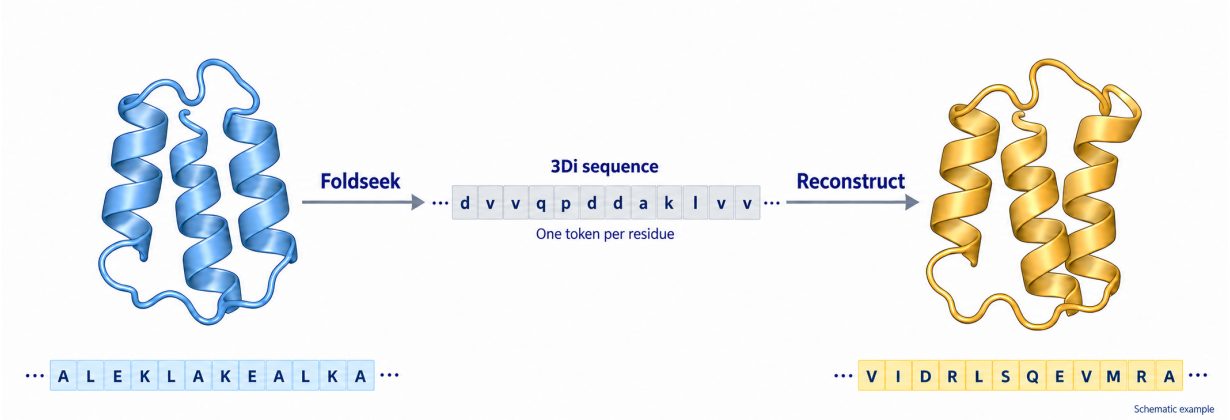


Figure 5. Representing a three-dimensional fold as a one-dimensional sequence for inverse folding. Foldseek encodes a protein backbone into a sequence of 3Di structural tokens with one token per residue, from which a new amino-acid sequence compatible with the same fold can be reconstructed.

This two-stage split offers a natural way to adapt OmegAMP for binder design by conditioning its sequence generation on a representation of the desired shape. Whereas ProteinMPNN represents that shape as a three-dimensional graph of the structure's geometry (Dauparas et al., 2022), OmegAMP operates on a sequence of residue features and therefore benefits from a sequence-based representation of structure. This raises the question of how to represent a three-dimensional fold as a one-dimensional sequence that still retains enough geometric information for inverse folding. Prior work on ProstT5 (Heinzinger et al., 2024) showed that this can be done using the 3Di structural alphabet, which encodes a protein's structure as a sequence of tokens with one token per residue (van Kempen et al., 2024), and training a protein language model to translate that structural sequence into an amino-acid sequence. Although ProstT5 achieved lower structural fidelity than ProteinMPNN in the reported comparison (Heinzinger et al., 2024), its results demonstrate that this compact representation retains information useful for inverse folding and can be aligned directly with the amino-acid sequence in OmegAMP.

Even with biochemical or structural conditioning, generated sequences are not guaranteed to function as intended. OmegAMP therefore filters its candidates with classifiers that predict antimicrobial activity (Soares et al., 2026), while in binder design we use structure-prediction models to assess both the shape a candidate binder adopts and how the binder and target look together. To predict how a protein's amino-acid sequence folds into a three-dimensional structure, AlphaFold2 combines the sequence with evolutionary information from multiple sequence alignments of related proteins, iteratively updating residue and residue-pair representations to predict coordinates (Jumper et al., 2021). ESMFold instead derives sequence representations from a pre-trained protein language model and passes them directly to a folding network, avoiding the slow alignment search for every prediction (Lin et al., 2023). Because binder assessment also requires folding the binder and target together to model their relative arrangement, both approaches have been extended to complexes. AlphaFold-Multimer adapts AlphaFold2 to jointly predict interacting proteins (Evans et al., 2022), while AlphaFold3 uses a diffusion module to denoise atomic coordinates for complexes of proteins, nucleic acids, and small molecules (Abramson et al., 2024). Similarly, ESMFold2 extends the language-model approach to predict complexes without multiple sequence alignments (Candido et al., 2026). We choose ESMFold2 for our pipeline because its alignment-free mode supports screening many generated candidates, for which informative evolutionary alignments may be unavailable, providing predicted structures and confidence scores to assess both binder shape recovery and placement against the target.

3 Method

3.1 Overview

Our overall goal is to develop a system for generating binders with control over their biochemical properties inspired by OmegAMP's approach. The central challenge is to extend OmegAMP's control over sequence generation so that the produced sequences not only have the desired biochemical properties but also the ability to bind a chosen target. Since binding requires the binder to adopt an appropriate shape, or scaffold, complementary to the target, this translates into a structural constraint on the sequences we generate. Our central task is therefore to train a generative model that combines OmegAMP's biochemical property control with structural guidance.

A natural approach would be to extend OmegAMP's model directly. By modifying its architecture and retraining it on suitable data, we would teach it to generate sequences of amino acids that both have the desired properties and fold into the desired structure. But this would require learning not only how to reconcile biochemical property control with structural requirements, but also how to identify a suitable scaffold for the target. Such an approach would entangle the task of determining a suitable binder structure with learning to generate sequences that realise it. Instead we separate out the task of finding the initial scaffold and choose to train a generative model that treats the scaffold as part of the design specification alongside the desired biochemical properties. This allows us to obtain initial scaffolds from models beyond our feasible training scale while retaining control over which structural features constrain generation. Scaffolds can be selected or revised, and individual regions left unspecified, to explore candidate binders with variation in both sequence and local structure.

Although our model treats the scaffold as given, the overall system should be able to generate binder candidates given a target rather than a scaffold. We therefore first obtain an initial scaffold for the target using an off-the-shelf model. Our generative model then uses this scaffold and the desired biochemical properties to guide sequence generation. As in OmegAMP, conditioning is intended to concentrate generation on promising candidates, but it does not establish the desired biological activity of every generated sequence. We consequently follow generation with a filtering stage, using structure prediction of the candidates' complexes with the target to assess their likelihood of binding.

3.2 Modeling the binding problem

Having described the general approach, we now describe in detail how the requirement to bind a target becomes an input supplied to our generative model. We begin by clarifying how we model the problem through providing the initial scaffold, then specify the structural and biochemical information we supply, and finally explain how we represent it for the model.

Fundamentally, given a target x, the goal of binder design is to generate a compatible binder $y \sim p(y \mid x)$. A direct approach would train a model to generate binder sequences from the target itself, effectively learning a mapping $F : X \to \mathcal{P}(Y)$, where $F(x)$ denotes the set of binders compatible with target x. But whether a binder is compatible with a target depends substantially on the structure and chemistry of the interface they jointly form. Shape determines how their surfaces can come into contact, while chemistry determines the interactions those contacts can support. This suggests decomposing the task into two stages. Rather than asking a model to generate a binder directly from the target, we can first determine a descriptor c of the intended interface and use that descriptor to generate a binder consistent with it.

$$x \xrightarrow[\text{derive the descriptor}]{H} c \xrightarrow[\text{generate a binder}]{G} y.$$

The advantage of this decomposition is that the intermediate specification is expressed in structural and chemical terms. It can therefore be supplied or revised using expert knowledge or a separate computational method, rather than having to be determined by the sequence generator itself. It describes the intended interaction without prescribing a complete binder, leaving the generator to determine a realization consistent with that specification.

The descriptor can be formulated from either side of the intended interface. A target-side description would characterize the surface that a binder must complement, whereas a binder-side description would specify the structural and chemical features proposed to provide that complementarity. These are different ways of expressing the constraints imposed by the intended interaction: one describes what the binder must interact with, while the other describes what the binder should present. The binder-side formulation is particularly convenient for design because it expresses requirements directly in terms of the molecule we wish to generate. We can revise its proposed characteristics while keeping the target fixed.

This choice nevertheless requires care: the descriptor supplies a desired binder shape without explicitly representing the target interactions that may influence it. Our generator is trained to realize binder specifications in isolation, so it is asked to produce sequences that adopt the desired shape without relying on those interactions. This imposes a more restrictive design objective than binding alone: rather than allowing the target to induce the desired conformation, we seek binders that already adopt approximately that conformation before binding. We further rely on them retaining it in the complex, even though the generator receives no explicit information about the target. We nevertheless consider this a reasonable simplifying assumption for our minimal pipeline, motivated partly by our initial desire to explore binder-conditioned approaches, and partly by experimental evidence of preorganization in designed minibinders (Cao et al., 2022).

3.2.1 Supplying structural information

The descriptor c must express both the biochemical requirements on a candidate binder and the structural characteristics it should realize. So far, we have referred to the binder's shape without specifying how precisely this structural component should be represented. The most detailed option would identify every atom and its three-dimensional position, as in a PDB file. That would overspecify the design problem: residue identities would already be fixed along with the geometry, leaving no sequence choices to make under the biochemical constraints. Instead, we describe the structure through the backbone, the continuous main chain linking the amino acids. Since amino-acid types are distinguished by their side chains, specifying only the backbone leaves residue identities open.

We must also choose how to represent the backbone for the model. Since we are adapting OmegAMP, we base this choice on how it represents peptides. OmegAMP represents a peptide as an ordered sequence of amino-acid feature vectors, each constructed from biochemical properties of the corresponding amino acid. Analogously, we seek a per-residue description of the backbone grounded in geometry. One fitting description, which we adopt, is the structural alphabet introduced by Foldseek, which represents an amino acid chain as a sequence of structural tokens.

These structural tokens, called 3Di, each describe the spatial relationship between a residue and a selected neighbouring residue in the folded structure. For each residue, Foldseek selects the residue with the nearest virtual center, a point derived from its backbone geometry. It then computes features describing the relative geometry of the two residues' backbone segments, including angles, distance, and separation along the sequence. A pretrained encoder maps these features into a two-dimensional latent space, where they are quantized into 20 discrete structural states, yielding a token at each residue position. Each state also has a corresponding two-dimensional vector in this latent space, which we use to represent the token in our model. Because the selected neighbour can be distant along the sequence, these tokens capture how different parts of the backbone are arranged in the folded structure (van Kempen et al., 2024).

We use these tokens to define the structural component of the descriptor c. More concretely, we specify the structure a generated binder should adopt through the sequence of 3Di tokens. Intuitively, the sequence describes the desired backbone in residue order, expressing the structural characteristics one residue at a time. Importantly, the provided sequence is allowed to be partial, so the specified 3Di tokens do not determine the binder's length, as unspecified positions allow but do not imply additional structurally unconstrained residues. The structural component of c is a sequence $s \in (\mathcal{A}_{\text{3Di}} \cup \{\perp\})^*$, where $\mathcal{A}_{\text{3Di}}$ is the 20-state structural alphabet and $\perp$ denotes an unspecified token.

3.2.2 Supplying biochemical information

Unlike the structural component, which is local in nature as it describes individual residue positions, the biochemical component of the descriptor c should be treated as a global specification, characterizing the binder as a whole. In specifying this component, we follow OmegAMP, using the desired net charge and mean hydrophobicity of the binder as a simple set of global properties to steer. Here, net charge describes the balance of its positive and negative electrical charges, while mean hydrophobicity describes the average tendency of its side chains to favour nonpolar environments over water.

We adopt an approach analogous to that used in OmegAMP to quantify hydrophobicity and net charge (Müller et al., 2017; Soares et al., 2026). For hydrophobicity, we use the Eisenberg consensus scale, which assigns a hydrophobicity score to each amino-acid type. The 20 values are normalized to have a mean of zero and a standard deviation of one, with higher scores indicating greater hydrophobicity (Eisenberg et al., 1984). To characterize the binder as a whole, we average these scores across all residues in its sequence. For net charge, we estimate the binder's total electrical charge by summing the positive and negative contributions of its side chains and the two ends of the chain. These contributions offset one another, so a binder can contain several charged residues while having little net charge overall. Because the contributions depend on pH, we use pH 7 as a fixed reference for all sequences.

Finally, next to charge and hydrophobicity, our descriptor c also provides the desired length of the binder as a further nonbiochemical, albeit global, property. In total, we end up with a set of three properties whose desired values we denote by ℓ, q, and h for length, charge, and hydrophobicity, respectively. Formally, we end up with a biochemical specification $b = (\ell, q, h)$. As with the structural component, this specification may be partial.

3.2.3 The generation task

Having specified both components of the descriptor, we can return to the decomposition we started from. We split binder design into deriving a descriptor for the target and generating a binder consistent with it. With the descriptor now fully specified, we turn to the second step, the generator G, whose role is to learn the relationship $c \xrightarrow{G} y$. In particular, we describe how G is realized as a generative model over amino-acid sequences, how these sequences are represented, and how the descriptor c enters to steer the generation.

To formulate the generative task we must first describe the form of what is being generated. Following OmegAMP (Soares et al., 2026), we represent the generated sequence as an ordered collection of amino-acid feature vectors $\mathbf{r}_a \in \mathbb{R}^5$, with five dimensions encoding biochemical characteristics of the corresponding residue: hydrophobicity (modified Wimley–White), isoelectric point, α-helix propensity (Levitt), transmembrane propensity, and the amino acid selectivity index (AASI). In our case, however, we extend this representation to include structural information as well. We use this joint generation of both sequence and structure to encourage the model to learn the relationship between amino-acid sequences and structural features. Consequently each feature vector also contains two coordinates $\mathbf{r}_s \in \mathbb{R}^2$ representing the corresponding residue's 3Di state in Foldseek's two-dimensional latent space. Concatenating these components gives a joint residue representation $\mathbf{r} = (\mathbf{r}_a, \mathbf{r}_s) \in \mathbb{R}^7$, so a binder of length L is represented by an ordered sequence of L such vectors.

To give binders of different lengths the same computational shape, we arrange these sequences of residue vectors as columns of a matrix with a fixed number $M \geq L$ of columns. Each binder occupies L consecutive positions in sequence order, with padding filling any positions before and after it. To clearly delineate residues from padding, we add a binary residue-presence dimension. A residue is then represented by $\widetilde{\mathbf{r}} = (\mathbf{r}_a, 1, \mathbf{r}_s) \in \mathbb{R}^8$. These eight-dimensional vectors form the columns of the matrix $y_0 \in \mathbb{R}^{8\times M}$, with k padding columns before the first residue:

$$(y_0)_{:,i} = \begin{cases} \widetilde{\mathbf{r}}_{i-k}, & k < i \leq k + L, \\ \mathbf{0}_8, & \text{otherwise}, \end{cases} \qquad 1 \leq i \leq M.$$

The descriptor c is structured to align directly with this matrix representation of y_0. We present s to the model as a matrix with the same number of columns as y_0. As such, the structural characteristics at position n in s correspond to the desired structural features at position n in y_0. Not only do their positions match; each supplied 3Di state is encoded using the same two-dimensional coordinates as the structural component of a residue vector and placed in the corresponding column, yielding a shared positional and numerical representation. The biochemical representation also aligns with the conditioning in the properties it describes, through features related to charge and hydrophobicity.

Because both components of the descriptor may be partial, we augment their numerical values with binary flags indicating what is supplied, setting unsupplied values and coordinates to zero. Each flag $a \in \{0, 1\}$ marks whether the corresponding property or 3Di token is supplied. When a 3Di token is not supplied ($a_i = 0$), the model still needs to distinguish residue positions whose structure was withheld from padding or structurally unconditioned positions. We therefore include a second flag $d_i \in \{0, 1\}$, setting $d_i = 1$ where a 3Di token is deliberately hidden ($a_i = 0, d_i = 1$) and $d_i = 0$ both where a token is supplied ($a_i = 1, d_i = 0$) and at padding or unconditioned positions ($a_i = 0, d_i = 0$). Writing (u_i, v_i) for the 3Di coordinates at column i, the encoded biochemical and structural conditions $B \in \mathbb{R}^6$ and $S \in \mathbb{R}^{4\times M}$ are

$$B = (\ell, a_\ell, q, a_q, h, a_h), \qquad S_{:,i} = (u_i, v_i, a_i, d_i)^\top.$$

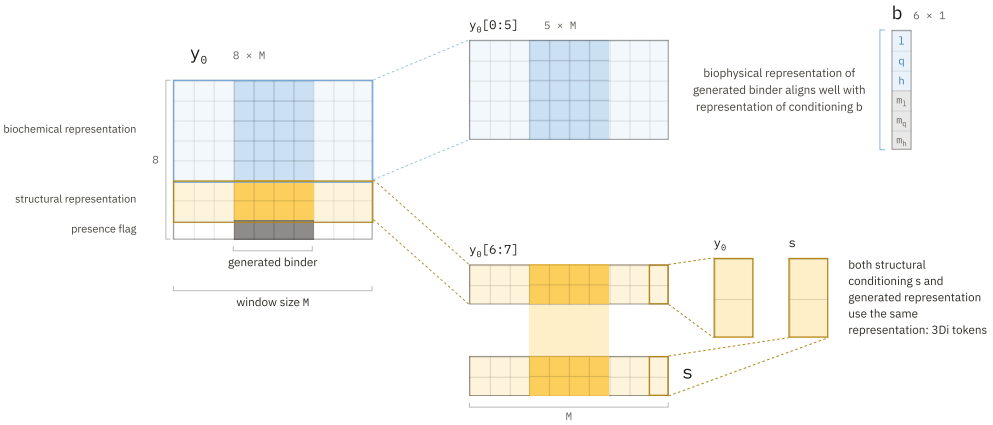


Figure 6. Correspondence between the generated binder representation and its structural and biochemical conditions.

Alongside the descriptor, we can supply parts of an existing sequence for the model to preserve. We represent this context through a template matrix $y_{\text{template}} \in \mathbb{R}^{8\times M}$ with the same layout as y_0, containing the supplied residue vectors and zeros at positions to be generated. An edit mask $m_e \in \{0,1\}^M$ distinguishes these positions, with a value of one requesting generation and zero requesting preservation of the template or fixed padding. For complete sequence generation, the template is zero and every position in the matrix is editable.

With the sequence representation and conditioning inputs in place, we realize G as a conditional diffusion model (Ho et al., 2020). During training, a denoising model D_θ learns to predict the clean matrix y_0 from a noisy version $y_t \in \mathbb{R}^{8\times M}$, given the diffusion step $t \in \{0,\ldots,T-1\}$, the encoded descriptor (S, B), and any supplied template and edit mask.

During generation, we start from Gaussian noise across the entire matrix and progressively remove it through reverse-diffusion steps guided by the model's predictions $\widehat{y}_0 \in \mathbb{R}^{8\times M}$. The template remains available as clean context throughout this process, but its supplied residues are not copied into the output. Instead, the model predicts every position and must learn to preserve the supplied context. The complete generative task therefore models $p(y_0 \mid c, y_{\text{template}}, m_e)$, as shown schematically in the figure below.

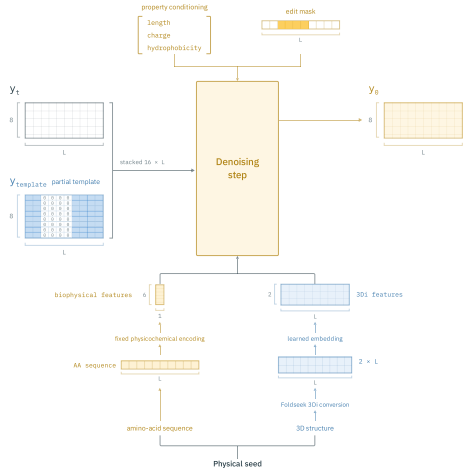


Figure 7. Generation loop for conditional binder sequence generation.

3.3 Model architecture

We realize the denoising model D_θ as a U-Net (Ronneberger et al., 2015) that jointly processes the amino acid and 3Di representations along the sequence. The supplied template context and edit mask are concatenated with the noisy representation as additional input channels before the initial convolution. The network first encodes the input through a downsampling path and then decodes the resulting features through an upsampling path, each comprising three stages. Skip connections pass detailed information directly between corresponding stages.

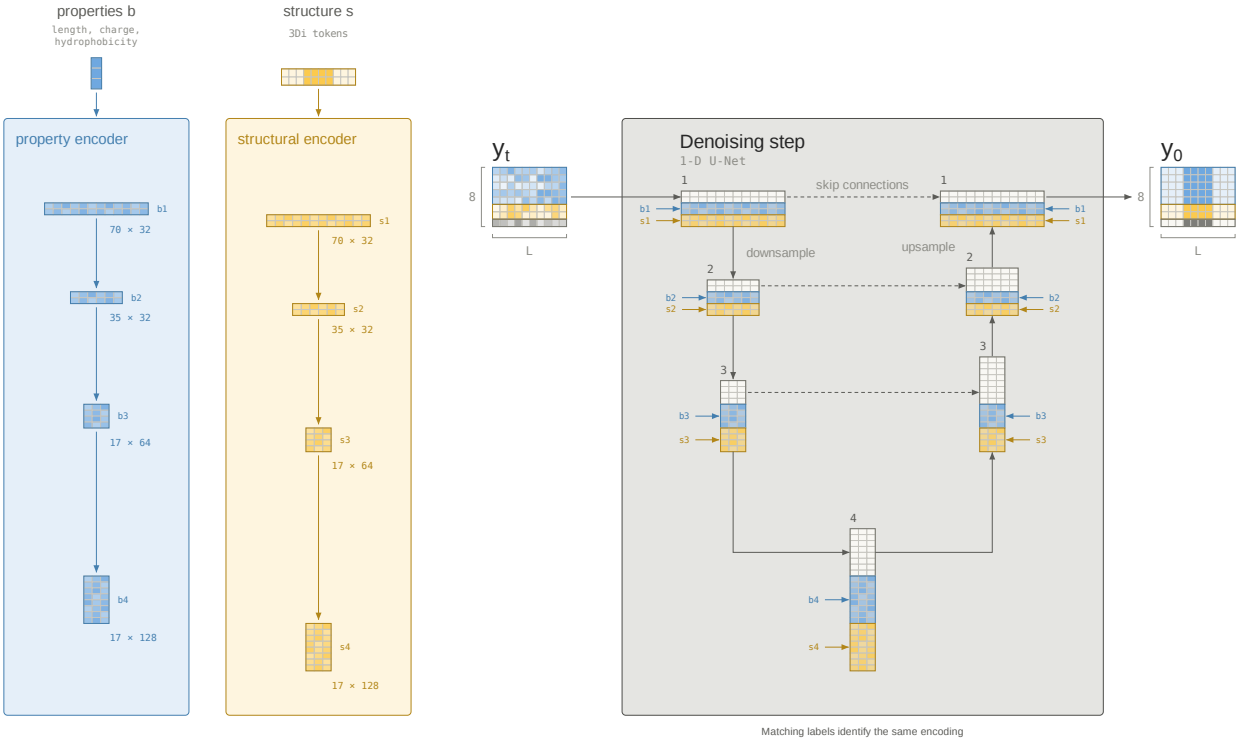


Figure 8. U-Net architecture with property and structural encoders and skip connections.

Both paths are guided by the descriptor through separate property and structural encoders. The property encoder expands the biochemical specification and its supplied flags into features arranged along the sequence. The structural encoder uses self-attention to combine information across the supplied 3Di positions ($a_i = 1$), allowing deliberately hidden positions ($a_i = 0, d_i = 1$) to attend to them through positional embeddings and infer contextual structural features. Positions with $a_i = d_i = 0$, or sequences with no supplied 3Di tokens, are masked to zero. Both encoders produce feature maps matching the U-Net's resolutions, which are concatenated with the denoising features inside the blocks of the downsampling path, bottleneck and upsampling path. This makes both conditions available throughout feature extraction and reconstruction. Each block also receives a sinusoidal encoding of the diffusion timestep t, allowing it to adapt its processing to the current denoising step.

Within these blocks, the denoising features are transformed together with the encoded conditions using one-dimensional convolutions. These mix information across channels and neighbouring sequence positions, producing an update that is added back to the incoming denoising features through a residual connection. Self-attention layers interspersed with the convolutional blocks extend this processing beyond local neighbourhoods, combining features across the sequence at the current resolution. This combination is repeated along both paths: during downsampling, it operates on increasingly coarse sequence representations, while during upsampling, it incorporates the detailed features supplied by the corresponding skip connections.

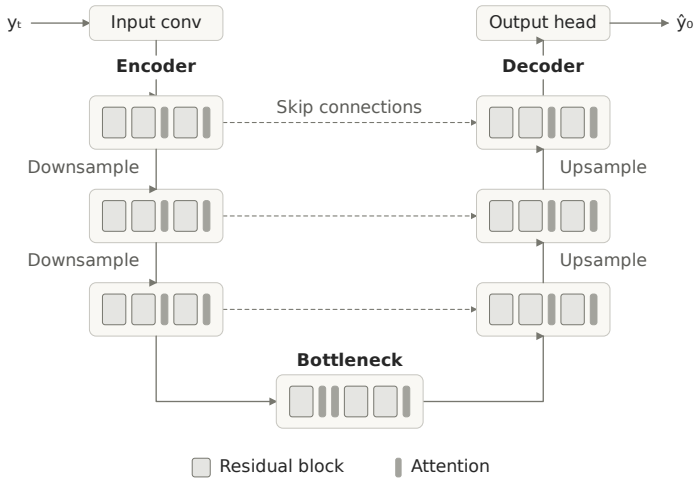


Figure 9. Layer organization of the denoising U-Net.

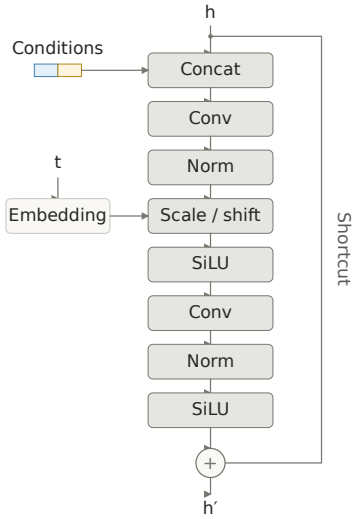


Figure 10. Residual convolutional block with descriptor and timestep conditioning.

To strengthen structural conditioning when structural guidance is enabled, the network predicts twice at each denoising step, comparing one prediction made with the requested 3Di input and one with that explicit input hidden. Following the principle of classifier-free guidance (Ho and Salimans, 2022), we amplify the difference between their amino-acid predictions to favour sequences consistent with the supplied structure. This guidance acts only on the amino-acid component of the denoising prediction, and structure prediction and interface scoring occur afterward without contributing to it. After denoising, the network's output is a continuous matrix, so we decode it into a sequence by assigning each position to the closest amino-acid or padding vector in Euclidean distance, using the five biochemical coordinates and the residue-presence coordinate. The two structural coordinates do not enter this assignment.

3.4 Training

Each training example starts as a complete protein. We derive a partial descriptor from it and train the model to reconstruct the complete protein representation from a noisy version, guided by that descriptor.

We originally intended for our training examples to come from Teddymer, a synthetic dataset that addresses the scarcity of binder–target data by treating interacting domains within predicted proteins as separate binding partners (Didi et al., 2026). In the original protein serving as the source of the synthetic pair, the two elements of the pair were connected through a linker. In our tests, removing this linker, and treating the pair as a target-binder complex, and refolding the ensuing structure tended to produce a different 3D outcome. This suggested that their contact could depend on that connection, raising doubts about treating them as examples of binding between independent molecules.

Instead, we exploit the fact that, strictly speaking, since our generator models the binder independently of the target, our training examples can in theory be monomers, not binders within a complex. This is doubly convenient, because geometric ML methods on proteins require many training examples which are scarce for binders. Hence, we actually opted to train on the ProstT5 dataset, which offers a much larger pool of approximately 17 million protein sequences paired with 3Di encodings derived from AlphaFold-predicted structures (Heinzinger et al., 2024). From these, we select complete sequences of 50–70 amino acids preserving the original data splits, giving 1.5 million training examples, 100 validation examples and 92 test examples.

To make the model useful with different amounts of prior information, we construct training examples with varying sequence, structural, and biochemical conditions. We first choose whether to start from the complete amino-acid sequence, a prefix, a suffix, a middle segment, or no sequence information. For the three partial-sequence modes, the initial segment's length is sampled uniformly between 35% and 80% of the protein's length. Within any nonempty starting segment, we then choose how many residues to hide, with every count from one to the entire segment equally likely. Rather than hiding these residues independently, we distribute them across contiguous spans. We sample the number of candidate spans from a geometric distribution with parameter $p = 0.5$, truncated between one and the number of residues to hide, favouring fewer spans. Each candidate span receives one hidden residue, and the remaining hidden residues are allocated among the spans using a symmetric Dirichlet–multinomial distribution with concentration $\alpha = 0.5$. A separate draw from the same distribution allocates the unhidden residues to the gaps between and around the spans. This allows uneven span lengths and gaps, with adjacent spans merging whenever their separating gap is zero.

We next determine which 3Di tokens to supply. In examples that retain part of the amino-acid sequence, positions with supplied amino acids always retain their structural information, while each contiguous missing sequence region, including any omitted terminal region, independently retains its 3Di information with probability 0.5. In examples starting without any amino-acid context, we instead choose whether to provide structural information, biochemical properties, both, or neither. To train the model to infer missing structural context, we then mask a random subset of any 3Di tokens retained at editable residue positions, sampling the number of kept tokens uniformly from zero to all available tokens and grouping them into contiguous spans with the same geometric and Dirichlet–multinomial distribution. Positions that retain their 3Di tokens have $a_i = 1$ and $d_i = 0$, positions masked in this step are marked as deliberately hidden with $a_i = 0$ and $d_i = 1$, and padding or regions whose 3Di was omitted earlier have $a_i = 0$ and $d_i = 0$.

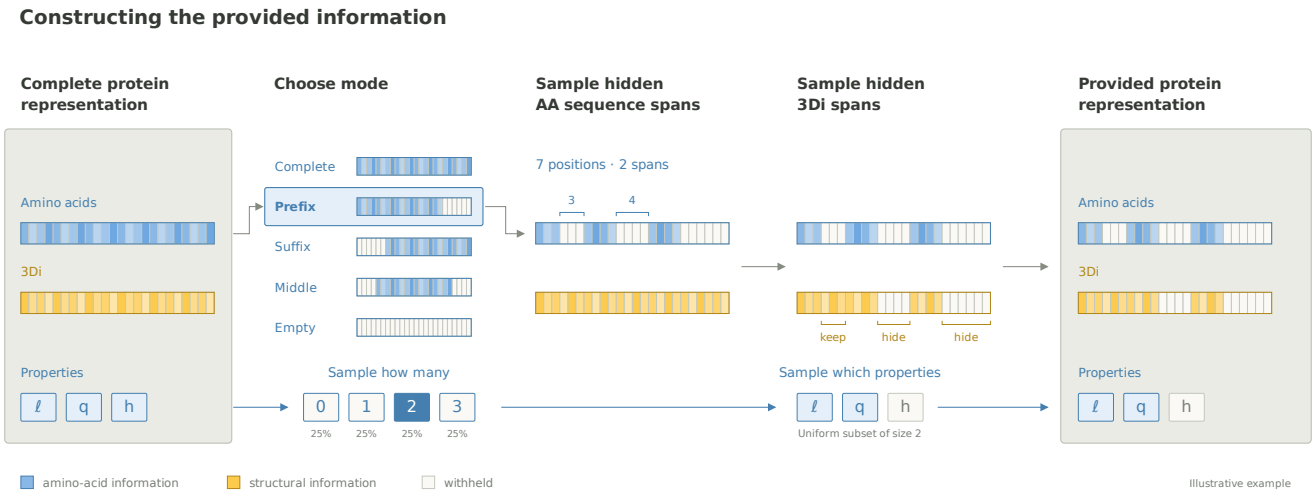


Figure 11. Complete protein representation, hidden span and property sampling, and provided protein representation.

Finally, we select which global biochemical properties to supply from length, net charge, and mean hydrophobicity. For sequence-editing examples, we assign equal probability to supplying zero, one, two, or three properties and uniformly select a subset of that size. Whenever properties are included for examples without amino-acid context, their number is sampled uniformly from one to three, ensuring that the selected conditioning mode supplies at least one property.

Using these examples, we train a conditional diffusion model in the continuous space of residue features. For each example, we sample one of 1,000 diffusion timesteps uniformly and mix its clean matrix y_0 with Gaussian noise to obtain y_t. Given y_t, the timestep and the partial descriptor, the model predicts the clean matrix $\widehat{y}_0$. We minimize mean squared reconstruction error across all eight channels, averaging the errors at residue and padding positions separately and giving the two groups equal weight. Training uses batches of 1,024 examples and bfloat16 mixed precision. Across successive training stages, we perform approximately 300,000 updates, corresponding to about 307 million presentations of training examples. We organize these stages to first learn sequence–structure relationships, then introduce property control and finally learn to complete partial sequences. For the first 150,000 steps, we provide only 3Di information, in varying amounts, and ask the model to generate an amino-acid sequence consistent with it. In the next 50,000 steps, we introduce conditioning on the biochemical properties. Finally, during the last 100,000 steps, half the examples continue this task, while the other half also supply partial amino-acid sequences for the model to complete. During inpainting, we extend the reconstruction loss to supplied residues as well, teaching the model to preserve them.

3.5 End-to-end pipeline

We have now described a model that is capable of generating amino acid sequences guided by the desired shape and biochemical properties. This is in itself insufficient on its own to design a binder from the perspective of the user. This brings us to the two unresolved parts of the pipeline. We still need to obtain a shape suitable for binding the chosen target and assess whether the generated sequences are likely to bind it by predicting their structures in complex with the target.

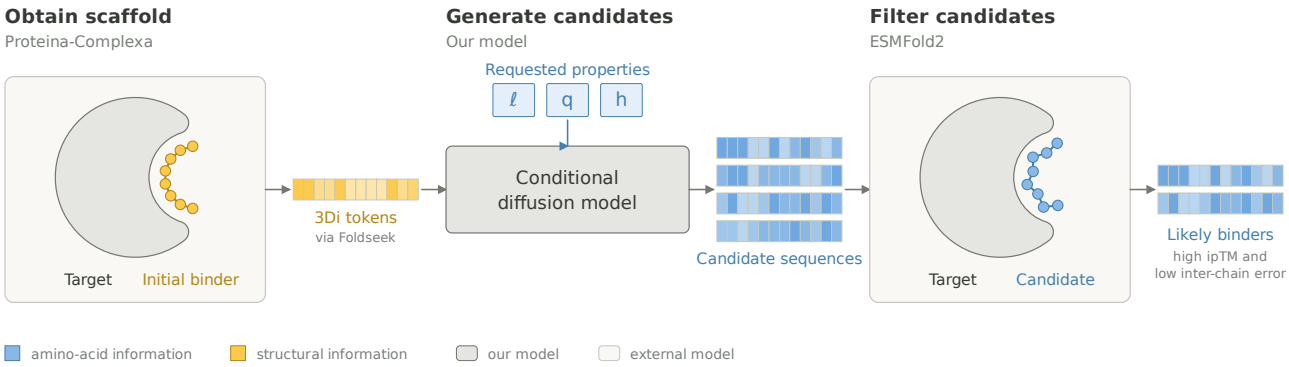


Figure 12. End-to-end pipeline, from obtaining a scaffold for the target to filtering the generated candidates.

For the first part, we obtain the shape from Proteina-Complexa, an off-the-shelf model much larger than ours that proposes an initial binder for the given target (Didi et al., 2026). We encode the backbone of this binder as a sequence of 3Di tokens, which forms the structural component s of the descriptor. Because the tokens are computed from the binder chain alone, they describe the binder in isolation, as in our training data. Together with the biochemical properties requested by the user, these tokens make up the descriptor c that we supply to our model. Positions that we do not wish to constrain can be left unspecified. For positions that we do not wish to regenerate, we supply the original residues as template context and ask the model to preserve them.

For the second part, we predict the structure of each generated candidate together with the target using ESMFold2, a structure prediction model that also handles complexes (Candido et al., 2026). Our generator never sees the target, so this is the only stage that checks the generated candidates against it. ESMFold2 reports an ipTM score, which measures confidence in how the two proteins are arranged relative to each other (Evans et al., 2022), and a predicted aligned error between residues (Jumper et al., 2021), which we average over pairs of residues from different proteins. We retain the candidates with a high ipTM and a low inter-chain error, since these are confidently predicted to form a complex with the target. These scores come from predicted structures, so they estimate the likelihood of binding without establishing it.

4 Evaluation

4.1 Unconditional evaluation

We start by evaluating whether our diffusion model can generate plausible protein sequences. We make every position in the matrix editable, provide no partial sequence, leave all 3Di positions unspecified, and specify no biochemical properties, thereby sampling from the learned marginal distribution $p(y)$. To do this evaluation we generate 1,024 sequences under these conditions and evaluate them against a background reference set of 92 proteins from the excluded ProstT5 test-set.

We first assess whether decoding produces well-formed and novel sequences. An output is considered valid if it contains exactly one nonempty, contiguous block of residues, with padding allowed only before or after it. Sequence validity is the proportion of outputs satisfying this criterion, while exact novelty is the proportion of valid outputs whose complete sequence does not appear in the training set. Of the 1,024 outputs, 1,022 were valid (99.8%), and every valid sequence was absent from the training set. The two invalid outputs contained internal padding and were excluded from the remaining analyses.

Second, we examine how strongly the valid generated sequences differ from one another and from the training data. We do this through three metrics based on Levenshtein distance, which measures the number of substitutions, insertions, and deletions needed to transform one sequence into another. We normalize each distance by the length of the longer sequence. To determine whether unique sequences are generated, we compute the proportion of distinct outputs, defining two sequences as distinct when their Levenshtein distance is greater than zero. To quantify overall diversity, we compute the mean normalized Levenshtein distance between generated-sequence pairs. Finally, to detect near-copies of the training data, we compute the normalized Levenshtein distance from each generated sequence to its closest training-set sequence. All valid outputs were distinct. Their mean pairwise normalized distance was 0.838, compared with 0.830 among the held-out proteins, and their mean distance to the closest training sequence was 0.640.

To assess whether the generated sequences resemble plausible proteins beyond simple sequence statistics, we evaluate them using ESMC-6B, a protein language model trained on large collections of natural protein sequences (Candido et al., 2026). Specifically, we compute pseudo-perplexity, the masked-language-model analogue of perplexity, which quantifies how surprising a sequence is under ESMC (Salazar et al., 2020). Since pseudo-perplexity has no absolute validity threshold, we compare its distribution for generated sequences with those of held-out ProstT5 proteins and PPIRef binders (Bushuiev et al., 2024). Median pseudo-perplexity was 17.02 for generated sequences, compared with 5.23 for ProstT5 and 2.76 for PPIRef. To explore this gap, we also evaluated shuffled sequences. The reference proteins scored substantially worse after shuffling, whereas generated scores barely changed and were similar to those of the shuffled references. This suggests that our diffusion model captures amino-acid composition better than the dependencies between residues.

Finally, we compare the generated sequences with held-out ProstT5 proteins at the distribution level. For sequence length, net charge, and hydrophobicity, we use the Kolmogorov–Smirnov statistic, which measures the largest difference between the cumulative distributions of each property. We compare their overall amino-acid compositions using Jensen–Shannon divergence, which measures the difference between the corresponding 20 amino-acid frequency distributions. In both cases, zero indicates identical distributions, while larger values indicate greater disagreement. Generated sequences were shorter on average, at 55.48 residues compared with 61.13 in the held-out set, and the length KS statistic was 0.468. Net charge and hydrophobicity showed smaller distributional differences, with KS statistics of 0.123 and 0.098, respectively. Overall amino-acid composition was also close under Jensen–Shannon divergence, at 0.00128 bits.

4.2 De novo biochemical property steering

To isolate the model's ability to follow the biochemical specification b, we supply only this component of the descriptor c. We again set the template to zero and make every position in the matrix editable, while leaving the structural component s unspecified.

Following OmegAMP, we uniformly sample complete biochemical specifications from the training set and divide them evenly among four conditioning settings: length only, net charge only, hydrophobicity only, and all three properties jointly. Sampling complete specifications before choosing which properties to supply ensures that the jointly specified values preserve the property correlations observed in the training data. For each generated sequence, we recompute its properties and measure their mean absolute error (MAE) relative to the supplied values, both directly and normalized by the corresponding training-set standard deviation.

We generated 1,000 sequences per setting. All outputs were valid, and the requested length was matched exactly in 99.8% of cases where length was specified. Charge and hydrophobicity also closely followed the supplied values, with similar errors when requested individually or together.

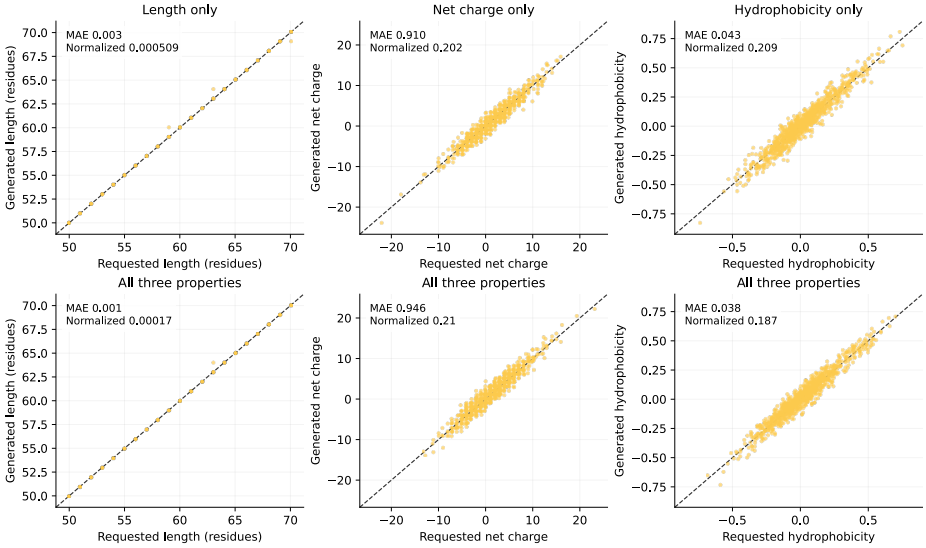


Figure 13. Requested and generated properties under single-property and joint conditioning

The plots above show how closely generated properties match the requested values under individual conditioning in the top row and joint conditioning in the bottom row. Dashed lines indicate an exact match. To put these steering errors in context, the figure below compares our results with the published single-property and joint-conditioning results from OmegAMP. Because the models were trained on different datasets, we normalize each model's errors by the corresponding property's standard deviation in its own training set.

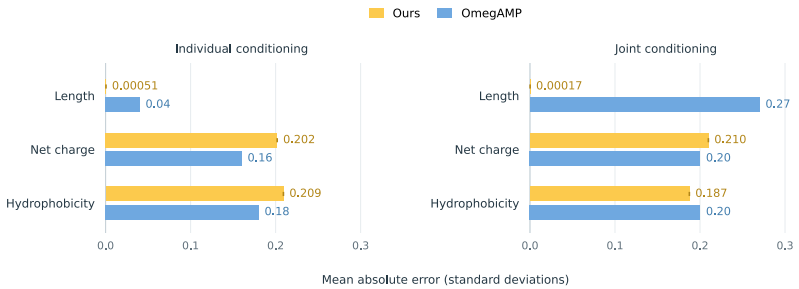


Figure 14. Property steering errors for our model and OmegAMP, comparing individual and joint conditioning in training-set standard deviations

4.3 3Di token steering

We next test whether the model can design sequences for a specified structural scaffold. For each protein in the held-out ProstT5 test set, we supply its complete 3Di representation as the structural component s, set the template to zero, and make every position in the matrix editable. We leave the biochemical specification b unspecified, so generation is conditioned only on s.

We generated ten candidate sequences per test protein with standard sampling and ten with structural guidance at a scale of 5, using the procedure described in the method section. We predicted all structures with ESMFold2-Fast (1 trunk, 3 loops, 20 sampling steps) and, for each protein in each sampling setting, selected the candidate with the highest mean pLDDT. We assessed the selected candidates' similarity to predictions for the corresponding original sequences using TM-score (Zhang and Skolnick, 2004). Mean TM-score was 0.462 with standard sampling and 0.495 with stronger guidance, on a scale where one denotes a perfect structural match.

We also examined how closely the predicted structures reproduced the supplied 3Di description. After converting the predictions back to 3Di, we measured exact token recovery and a soft similarity score that gives partial credit to similar states using Foldseek's substitution matrix (van Kempen et al., 2024). On average, standard sampling recovered 52.7% of the requested tokens and achieved 55.2% soft similarity. Stronger guidance raised these values to 61.9% and 65.0%. Processing the original sequences in the same way gave 62.6% exact recovery and 65.3% soft similarity. With stronger guidance and best-of-10 selection by pLDDT, the generated sequences therefore achieved similar mean 3Di agreement to the original-sequence refolds, despite reaching a mean TM-score of only 0.495. This gap indicates that matching the 3Di description, which is itself only partially preserved under refolding, does not fully pin down the global three-dimensional geometry.

4.4 Pipeline evaluation on experimental complexes

Having tested property and structural conditioning separately, we now evaluate them together within the design pipeline. Given a target protein, an initial target-compatible binder, and a requested biochemical specification, can the pipeline generate a modified binder that satisfies the requested properties while remaining compatible with the target?

Rather than generating initial scaffolds with Proteina-Complexa, where upstream failures would confound the benchmark, we start from 81 curated PPIRef complexes (Bushuiev et al., 2024) with binder constructs of 50–70 residues, using their experimentally resolved binder segments as the initial scaffolds. Before redesign, we predict the original complexes with ESMFold2-Fast and retain 45 with close binder-fold recovery, high binder and interface confidence, and low inter-chain predicted error. This restricts the evaluation to cases for which the prediction stage provides a confident baseline for the original binder.

We first consider the more demanding setting of complete sequence generation, supplying the binder's complete 3Di representation as s and original property values as b, with no template context. At first glance, requesting the original properties rather than a new profile may seem not to test property steering. Yet requesting arbitrary properties for a fixed scaffold could fail simply because no compatible sequence exists, whereas the original binder establishes that a target-compatible solution to c exists without supplying its amino-acid sequence to the model. This setup therefore tests whether generation can be steered toward a known-feasible specification, getting around the difficulty of designing a test suite of new property targets that are guaranteed to be solvable. The caveat is that this setup does not provide a picture of how far the biochemical properties can be varied on a 3Di-encoded scaffold that is held constant.

To do this evaluation, we generate 50 possible sequences for each binder in the curated set and predict each in complex with its target. Rather than imposing a fixed binary confidence cutoff across diverse complexes, we rank each binder's candidates using a weighted combination of confidence in the binder structure (pLDDT) and in the interface (ipTM), with a small penalty for normalized property errors, and compare the top-ranked candidates across the set through their mean scores. The selected candidates achieve property errors comparable to the de novo steering evaluation (length, charge and hydrophobicity MAEs of 0, 1.19 and 0.030, respectively), with exact length matching, somewhat higher charge error and lower hydrophobicity error. Their mean binder TM-score of 0.546 exceeds the commonly used threshold of 0.5 for overall fold similarity (Xu and Zhang, 2010). However, the selected candidates' mean ipTM was 0.574, leaving uncertainty about their compatibility with the target despite recovery of the overall binder fold.

We also evaluate an easier setting of inpainting, where we diversify a binder with part of its amino-acid sequence supplied as template context. We do this by masking 50% of the amino-acid positions in contiguous spans as described in the method section, while supplying the remaining residues in the template and retaining the complete descriptor c. We generate 20 completions per binder using different masks and apply the same selection score, selecting for each binder the highest-scoring completion that preserves the supplied residues. The selected candidates achieve a mean binder TM-score of 0.821 and mean ipTM of 0.743, improving both fold recovery and interface confidence compared with complete sequence generation despite screening only 20 candidates per binder rather than 50. They change approximately 80% of the masked residues while maintaining property accuracy comparable to the de novo steering evaluation (length, charge and hydrophobicity MAEs of 0, 1.03 and 0.029, respectively). The table below summarizes both settings alongside the refolded original binders.

Table 1. Mean scores for the selected candidate per complex, rather than across all generated candidates, alongside the refolded original binders.

Setting	Candidates	Binder TM-score	ipTM	Length MAE	Charge MAE	Hydro. MAE
Original binder refold	N/A	0.941	0.891	0	0	0
Complete generation	50	0.546	0.574	0	1.19	0.030
Inpainting, 50% masked	20	0.821	0.743	0	1.03	0.029

The numbers in the table reflect a particular balance in our selection score between structural confidence and property accuracy. If we penalize property errors more heavily when choosing among the generated candidates, we can select binders with smaller property errors at the cost of lower predicted interface confidence, as shown in the figure below. Overall, these results support inpainting followed by selection as a more practical mode of constrained redesign, which we illustrate on individual targets in the next section.

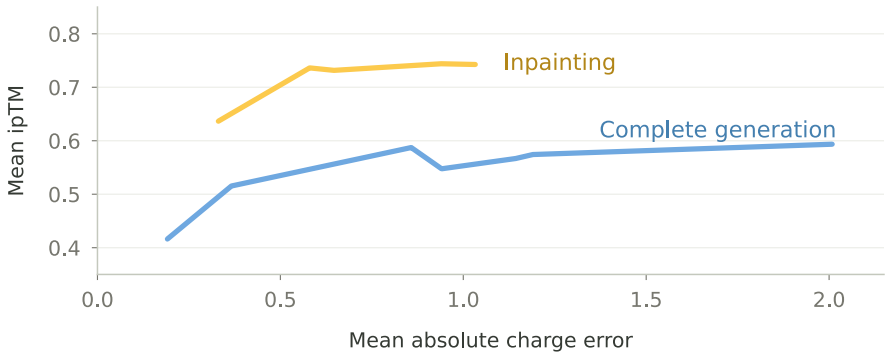


Figure 15. Trade-off between charge accuracy and predicted interface confidence across selection weights.

4.5 Design against medically relevant targets

To illustrate the complete pipeline starting from a target structure alone, we designed minibinders against three medically relevant proteins. The first is PD-L1, where blocking its interaction with the PD-1 receptor can help restore T-cell responses against cancer cells (Brahmer et al., 2012). The second is IL-7 receptor α (IL-7Rα), which supports lymphocyte survival and proliferation (Cao et al., 2022), so blocking its interaction with IL-7 could help limit autoimmune activity or leukemia growth (Barata et al., 2004; Hartgring et al., 2010). The third is the major birch pollen allergen Bet-v-1, where covering the surface recognized by IgE antibodies could interfere with the allergic response (Zettl et al., 2022).

For each target, we used Proteina-Complexa to generate 100 initial seed binders and selected five top-scoring scaffolds after refolding with ESMFold2-Fast. We first tested whether our model could generate entirely new sequences from a seed's 3Di scaffold and original biochemical properties alone. Screening 200 generated sequences per scaffold (1,000 per target) yielded candidates that matched the requested properties and formed confident predicted complexes for PD-L1 (binder pLDDT 80.9, ipTM 0.770, shown below) and IL-7Rα (binder pLDDT 81.1, ipTM 0.723).

For Bet-v-1, complete sequence generation produced a confident candidate (binder pLDDT 85.0, ipTM 0.780), but ESMFold2 predicted it to bind at a different site on the target than the original seed. The best candidate that remained at the intended binding site had lower confidence (binder pLDDT 65.9, ipTM 0.667). This illustrates a limitation of our descriptor, which specifies the binder's shape and biochemical properties without information about the target or where on it to bind. Inpainting the original Bet-v-1 binder instead yielded confident candidates at the intended site, suggesting that preserving part of the seed sequence helps constrain the binding location.

We next tested whether we could shift a seed's biochemical properties while maintaining predicted binding. Taking a promising scaffold for each target, we requested either an increase of 2 in net charge or an increase of 0.10 in mean hydrophobicity while keeping the other properties fixed. We screened 500 candidates per condition using complete sequence generation for PD-L1 and 200 candidates per condition using inpainting for IL-7Rα and Bet-v-1. Across all three targets, the selected variants retained confident complex predictions. Their net charge increased by approximately 2 as requested, whereas the hydrophobicity shifts had noticeably larger relative errors, ranging from 0.057 to 0.162, though we did not investigate this discrepancy further.

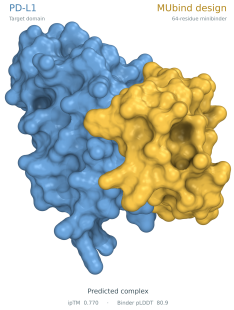


Figure 16. Predicted complex of PD-L1 (blue) with a 64-residue minibinder generated by MUbind (gold), shown as molecular surfaces. This full-sequence design was refolded with ESMFold2-Fast, giving a binder pLDDT of 80.9 and an ipTM of 0.770.

5 Discussion

We set out to extend OmegAMP's control over biochemical properties to binder design. Our results show that this control can be retained alongside useful structural guidance: generated sequences closely follow the requested properties, and given some filtering, generated candidates can yield reasonable recovery of the intended overall fold. The greater difficulty lies in the interaction with the target. In full sequence design, we do not reliably obtain candidates confidently predicted to form the intended complex, despite the control achieved over the binder's properties and overall shape. Inpainting is more successful, yielding such candidates while still allowing substantial changes to the original sequence.

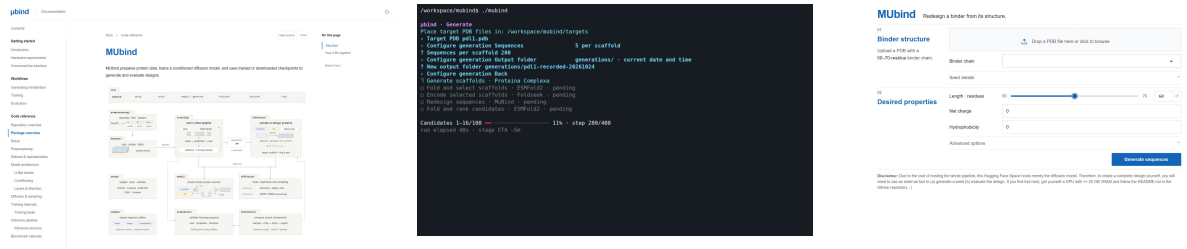
Several factors may contribute to this difficulty. First, although the model can recover the overall fold, its limited structural precision may partly reflect the compact 3Di representation, which does not specify the binder's exact geometry. Differences in local structure may be particularly important at the binding interface, where even small deviations can affect how well the binder fits the target. Second, our generator receives no information about the target itself. Although the supplied binder shape constrains the sequences it can propose, it does not communicate which amino acids would form favourable interactions with the target at particular interface positions. The model therefore cannot explicitly account for these interactions when choosing the sequence, whereas retaining part of the seed sequence during inpainting helps anchor the binding site and interface contacts. Finally, our unconditional ESMC-6B evaluation suggests that the model captures amino-acid composition more successfully than the dependencies between residues. Because net charge and mean hydrophobicity depend on which amino acids are present rather than how they are ordered, this limitation affects structural precision and binding far more than property control.

Despite these limitations, we have developed a pipeline that can generate minibinder candidates with the requested properties and select those confidently predicted to form the intended complex. The model's relatively compact size makes searching many candidates practical, so usefulness does not require every generated sequence to succeed. As our larger screens on individual targets suggest, cheap generation can partly offset low single-sample reliability by sampling more candidates, making the allocation of compute between candidate generation and structure prediction an important practical trade-off. Relying on predicted structures also makes it difficult to fully separate the generator's limitations from those of the evaluator, since even original sequences and experimental complexes often change under refolding. Overall, these results were obtained within the constraints we set at the outset of our project: retaining OmegAMP's U-Net architecture with limited modifications, using the binder's 3Di description as the sole structural input, and excluding target conditioning. These choices deliberately restricted the approaches we could explore, and although our experiments do not isolate their individual effects or rule out gains from better sequence modeling, we believe reliable binder generation will require revisiting these constraints rather than further refining the network alone.

Future work could build on inpainting by making the search adaptive, using feedback from predicted complexes to choose which regions to regenerate in successive rounds. This would allow target compatibility to guide further sequence changes rather than serving only as a final filtering criterion. More broadly, separating the design descriptor from the sequence generator provides a natural framework for testing which aspects of the target, the binder, and their intended interaction are actually needed to guide generation. For example, the descriptor could incorporate position-specific chemical requirements alongside the global biochemical specification, describing what the binder should present at its interface without prescribing exact amino-acid identities. Improving the model's ability to realise these requirements could involve structural representations more detailed than 3Di and alternative network architectures. Fine-tuning a larger pretrained structural model would offer another way to approach this task, building on existing sequence–structure knowledge rather than learning it from scratch. Alongside these methodological directions, the choice of controlled properties deserves further investigation. Our emphasis was on learning to control a selected set of properties, rather than systematically determining which are most relevant to binder design. Future work could therefore connect the choice of design requirements more closely to the intended application.

Appendix

We release our model and design pipeline in a GitHub repository named *MU*bind (our work is a combination of ideas from *Alpha*Fold and *Omeg*AMP, after all). The repository provides a Python package and an interactive command-line interface for binder design, alongside documentation for setup, generation, training, and evaluation. For quick experimentation without local installation, we also provide a web demo on Hugging Face Spaces, where users can sample sequences from the diffusion model under chosen biochemical and structural conditions. Links to the documentation and demo are available in the repository.



(1) Documentation (2) Command-line interface (3) Hugging Face

Figure 17. We provide documentation (1), a command-line interface (2), and a web demo hosted on Hugging Face (3).

GitHub repository

`https://github.com/B1RO/mubind`

Scan to open the MUbind repository.

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